Fiscal Year:	FY 2014	**Task Last Updated:**	FY 04/01/2014
PI Name:	Diddams, Scott A. Ph.D.		
Project Title:	Development of an Optical Microcomb System in Support of Fundamental Physics Research on the International Space Station		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Fundamental physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2011 Fundamental Physics NNH11ZTT001N
Start Date:	06/01/2012	**End Date:**	05/31/2017
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JPL
Contact Monitor:	Israelsson, Ulf	**Contact Phone:**	
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	None		
COI Name (Institution):	Papp, Scott (National Institute of Standards & Technology) Vahala, Kerry (California Institute of Technology)		
Grant/Contract No.:	NNH12AT81I		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Our research program will investigate and develop advanced microresonator optical frequency comb technology in support of planned European Space Agency (ESA) activities on the International Space Station (ISS). Optical frequency combs, which provide a dense set of absolute frequency reference lines, are an integral component of modern optical atomic clocks. They provide a link between optical and microwave signals and a means to access the stability of optical clocks at any wavelength. For example, a frequency comb will be required for the Space Optical Clock (SOC) experiment to achieve all of its fundamental scientific goals by providing the critical phase coherent optical or microwave link between the atomic frequency standard on the ISS and ground-based clocks. The same frequency comb technology will be a required component of other proposed ESA and NASA fundamental-physics clock experiments, as well as an enabling technology for ESA's planned Quantum Weak Equivalence Principle (QWEP) atom interferometry experiments. Microresonator combs (microcombs) can be lithographically patterned and fabricated on a single silicon chip with dimensions of a few square centimeters and therefore offer revolutionary advantages beyond any other conventional laser comb technology in terms of size, power consumption, and sensitivity to their environment. However, the full capability of a microcomb for precision optical frequency metrology has not been realized. The laboratory experiments we propose will, for the first time, characterize a microcomb's stability with respect to known optical standards, thus verifying its suitability to support the high-performance optical clocks envisioned for SOC and other space missions. Our microcomb system will be tested and interfaced with state-of-the-art optical clocks based on strontium and ytterbium atoms, which are the two candidate SOC platforms. Furthermore, the microcomb will have an output at 1064 nm and a countable microwave repetition rate in the 10-50 GHz range, which are compatible with the optical and microwave links required to achieve the fundamental physics goals of the SOC mission. Our research will develop advanced technology to enable full on-chip integration of a microcomb with essential components for time and frequency metrology including, high-Q silica microresonators, clock-signal coupling, comb self-referencing, and fiber v-groove end-couplers. These efforts will contribute to the long-term goal of realizing a single chip-scale device operating at the convergence of optics, frequency metrology, opto-mechanics, and microwave technology.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The best optical clocks now keep time with uncertainty measured in the 18th digit; however, their complexity constrains such remarkable precision to a research laboratory setting. Our research is focused on developing key components of optical clocks that will eventually enable their use outside the lab and in outer space. In particular, we aim to develop the compact “clockwork” that will enable the counting of optical cycles on the femtosecond time scale, as well as the synthesis of optical and microwave frequencies critical for transmission and comparison of multiple atomic clocks. Precision time keeping, as explored in our research, is anticipated to impact a wide variety of human activities both in Earth and space. Not only do clocks provide an accurate tool to analyze the inner workings of atoms and molecules, but they are critical to communication and navigation systems that we rely on every day. Examples include the global positioning system and cellular phone networks. Our research is an important step in developing the next generation clocks which will not only be more accurate, but also more compact and easily fabricated.
Task Progress:	In the past decade we have witnessed a revolution in time keeping with the introduction of clocks operating at optical frequencies. Atoms provide an outstanding absolute frequency reference for an optical clock, and precision comparisons between clock transitions of different atomic species can now be performed with fractional uncertainty below 1e-17. The remarkable precision of clock measurements provide a promising route to probing fundamental physical phenomena that include general and special relativity, local position invariance, Lorentz and CPT violation, possible variations of fundamental constants, and strongly correlated quantum states of matter. Measured differences from known physical law would have profound implications for our understanding of nature. Future experiments in search of new physics are under development; for enhanced sensitivity they leverage unique capabilities of microgravity and worldwide coverage to ground-based laboratories that are available on the International Space Station. However, nearly all these measurements require the comparison of at least two clocks that are spatially separated, and for most optical clocks under development the frequency of the atomic transitions is not ideally suited for long-range distribution in free space or optical fiber. Furthermore, optical clocks based on different atoms have transitions of interest spread across hundreds of terahertz, and an additional 5 orders of magnitude separate the optical from the microwave domain. Upcoming optical clock experiments on the ISS will use the same transformative technology that has made possible clock comparisons on the ground: The self-referenced optical frequency comb. Optical combs, with a dense set of absolute reference lines spanning more than an octave, permit the precise counting of individual cycles of light waves and measurement of time in units of femtoseconds (1e-15 s). Moreover, they can interface optical clocks both to microwave frequencies and to laser wavelengths that are optimized for space-to-ground transmission. Frequency comb technology now runs reliably in research laboratories, and advances in fiber lasers have led to robust, turn-key systems. However, these systems still remain relatively large, expensive, and sensitive scientific instruments. Support of future ISS optical clock missions requires the development of frequency comb technology that will be appropriate for space flight. Recently, a new class of frequency combs has emerged based upon chip-compatible microresonators. Here the comb generation is simply provided by parametric nonlinear optics, which can initiate with only mWs of pump power due to the high Q and small mode volume of microresonators. Such microresonator combs (microcombs) can be lithographically patterned and fabricated on a single silicon chip and therefore offer revolutionary advantages beyond any other conventional laser comb technology in terms of size, reliability, power consumption, integration, and sensitivity to their environment. Using standard techniques, it now appears feasible to fabricate an entire frequency comb system on a piece of silicon that is on the order of 10 cm2 and just a few hundred microns thick. Our research program is focused jointly on developing integrated microresonator technology for complete frequency combs systems, and on investigations of parametric comb generation to characterize the suitability of microcombs for precision optical frequency metrology. Team members at Caltech take the lead on device design, fabrication/integration and testing, while the NIST team is charged with demonstrating stabilization of the microcomb spectrum to optical clocks. In the second year of the program, our research activities were focused in two tasks: 1. The development of dispersion engineering techniques for fused-silica microresonators to facilitate broadband comb generation. The expansion of microcomb spectra towards octave or 2/3 octave spectra spans is being approached through

	engineering of resonator dispersion so as to flatten it over a wide range of wavelengths. Dispersion (both material and geometrical) set a bandwidth limit on comb operation unless control is introduced. The approach under investigation parallels methods commonly employed in optical fiber systems to balance silica material dispersion with a controllable geometrical dispersion. In the case of the disk resonators this is being done by shaping the contour of the dielectric boundary into a series of facets of predetermined angle and length. By modeling these structures we have shown that a net dispersion flattening is possible through careful facet design. Measurements of the dispersion at specific wavelengths agree well with the modeling and comb operation at spans of nearly 900nm and repetition rates of 30 GHz (a record span at this repetition rate) have been obtained. Current plans are to extend the measurement capability of dispersion towards longer wavelengths to further verify dispersion control using this approach. 2. Full optical stabilization of a microcomb, towards the demonstration of an interface with the Sr and Yb optical clocks. All-optical frequency control of a microcomb, including frequency division to the microwave domain, has been a crucial, yet unmet, challenge. During the past year, we reported the first achievement of this goal by demonstrating a functional optical clock based on stabilization of a microcomb to atomic Rb transitions. For this experiment, we generate a low-noise, continuously equidistant microcomb spectrum by use of an on-chip silica microresonator. The clock output is the ~33 GHz microcomb line spacing, which is electronically measurable, and a traceable, integer partition of the 3.5 THz frequency spacing of the Rb references. This experiment explored the basic features of a microcomb clock, and demonstrated that its fractional Allan deviation can reach 10-12 for 10,000 s of averaging. The noise component due to the microcomb itself is only 10-14 at one second, indicating that microcombs could support higher performance cold-atom optical references.
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